EE 210L: Digital Circuits and Logic Design Lab

Lab 6: Adders and Comparators

Name: _____ Date: _____

Partner(s): _____ Section: _____

Record logic levels, 0 or 1. Build the 74LS283 wiring table before you place the chip: its pin order is not sequential.

Part 5.1: Four-Bit Addition (74LS283)

Wiring table first. A1/B1/Σ1 are the least significant bit.

Bit	A pin	B pin	Σ pin
1, LSB			
2			
3			
4, MSB			

A (dec)	A (bin)	B (dec)	B (bin)	Σ4 Σ3 Σ2 Σ1	C4	Sum as decimal	4-bit result correct?
9	1001	6	0110				
9	1001	11	1011				

How does C4 tell you that a 4-bit sum has overflowed?

_____

_____

Part 5.2: Adder/Subtractor

Draw how the XOR gates and the control line connect to the 74LS283.

Figure 1: Adder/subtractor built around the 74LS283.

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Control	Operation	A	B	Σ4 Σ3 Σ2 Σ1	C4	Result as decimal	Expected
0	A + B	7	3				
1	A − B	7	3				
1	A − B	12	5				

What does C4 mean when the circuit is subtracting?

Part 5.3: Magnitude Comparator (74LS85)

Cascade inputs for standalone operation: pin 4 = _____ pin 2 = _____ pin 3 = _____

#	A (bin)	B (bin)	A>B out	A=B out	A<B out	Correct relation?
1						
2						
3						
4						
5						
6						

Include one pair with A > B, one with A < B, one with A = B, and one pair that differs only in the least significant bit.

Discussion and Conclusions

Summarize what you built and what it confirmed. Address at least: why a half adder cannot be used for any bit but the least significant, what ripple carry costs you as the adder gets wider, and how one chip manages to both add and subtract.

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